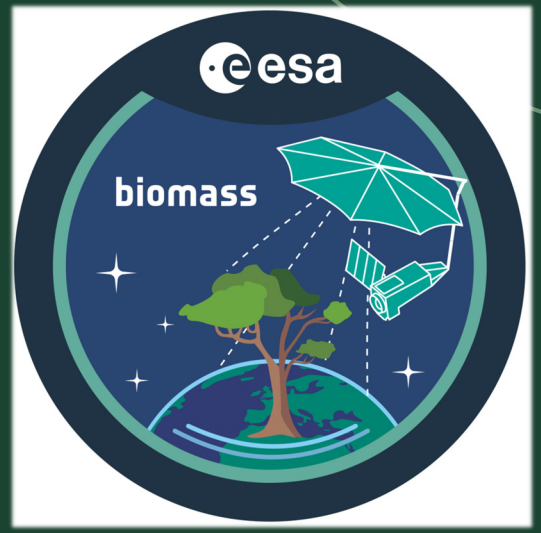
BRAVE

Biomass Radar for Assessing Vegetation and Ecosystems

AI-based Above Ground Biomass retrieval from ESA BIOMASS P-band SAR data

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Co-PIs: Emanuele Santi · Claudia Notarnicola



Project Objectives

A. P-, L-, C-band Sensitivity to AGB

- P-band penetrate the canopy and retrieve trunk and stem biomass
- Shorter wavelengths are more sensitive to crown biomass
- Integration with L-band (SAOCOM, NISAR) and C-band (Sentinel-1)

B. AI Retrieval Algorithm

- Define a refined AI algorithm (ANN / Random Forest) retrieving AGB from P-band
- Integrate L-band SAR (SAOCOM, NISAR) and C-band (Sentinel-1) when available

C. Comparison

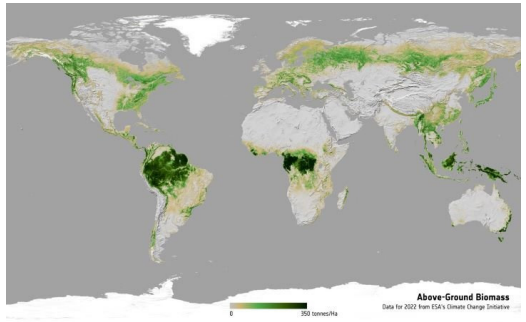
- Compare HydroGNSS Level 3 (FB) Forest Biomass product by using ESA BIOMASS SAR data.

Test Sites & Data

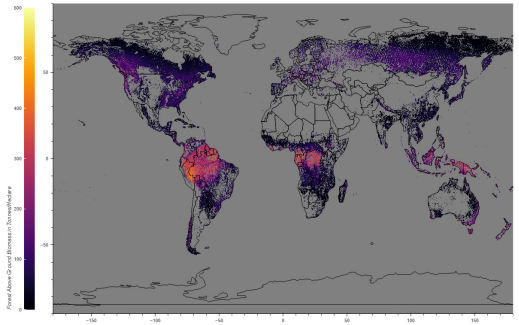
Test Sites: a) Gabon, Africa b) Manaus, Brazil

Date	Sat	Mode	Pass	Polarizations	Site
06/06/2025 17:18	BIOMASS	S1	D	HH, HV, VH, VV	Gabon – Africa
08/06/2025 16:49	SAO1A	DTWS2	D	HH, HV	Gabon – Africa
03/06/2025 04:50	Sentinel-1A	IW 80	D	VV, VH	Gabon – Africa
09/06/2025 09:56	BIOMASS	S1	A	HH, HV, VH, VV	Manaus – Brazil
11/06/2025 21:44	SAO1B	DTWS2	D	VV, VH	Manaus – Brazil
03/06/2025 09:47	Sentinel-1A	IW 83	D	VV, VH	Manaus – Brazil

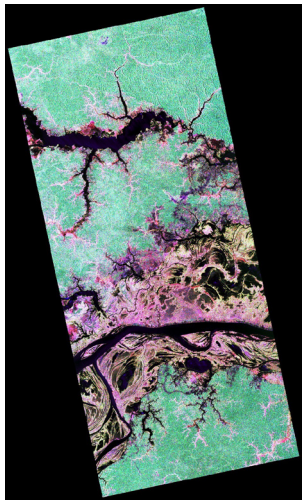
Forest data



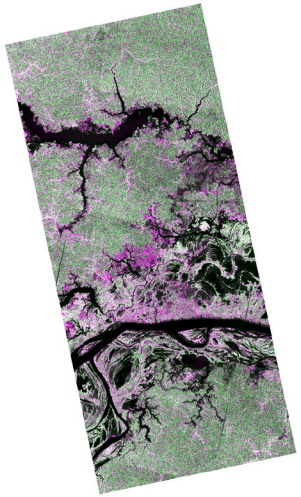
Biomass CCI, 2024



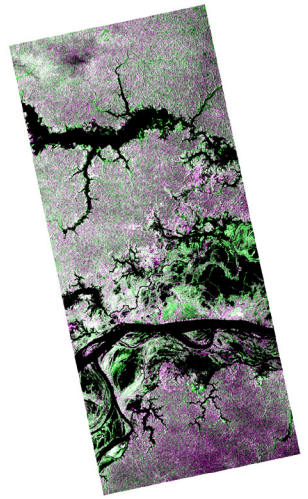
HydroGNSS L3 Forest Biomass (2026)



BIOMASS



SAOCOM



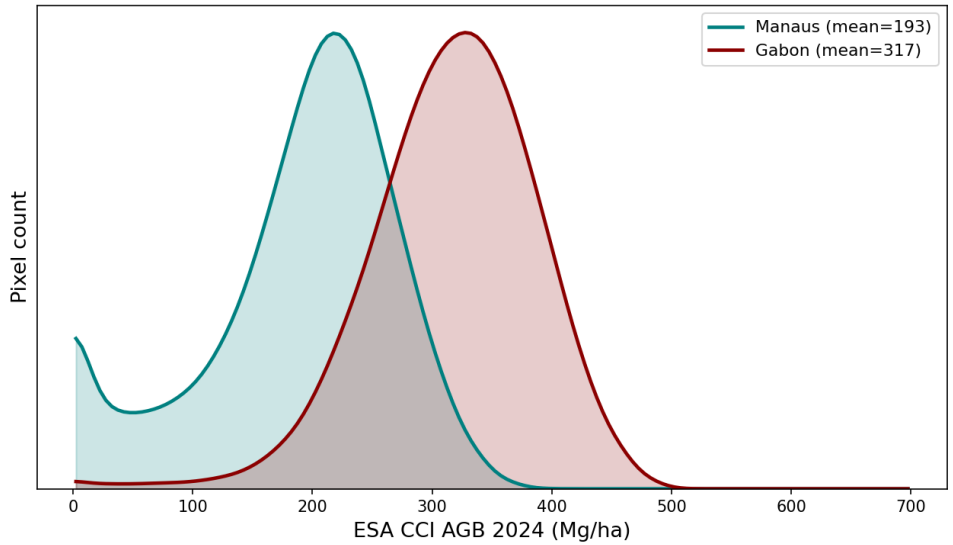
SENTINEL-1

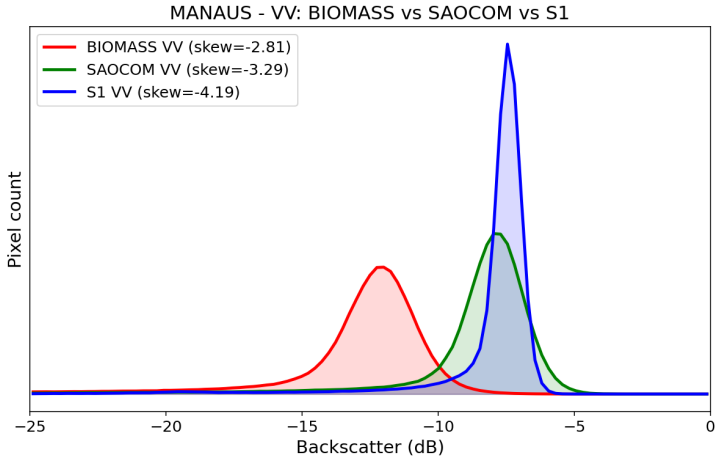
ESA Climate Change Initiative Biomass, 2024 (100 m)
ESA HydroGNSS L3 FB product 6 months, (25 km) – available from Nov 2026

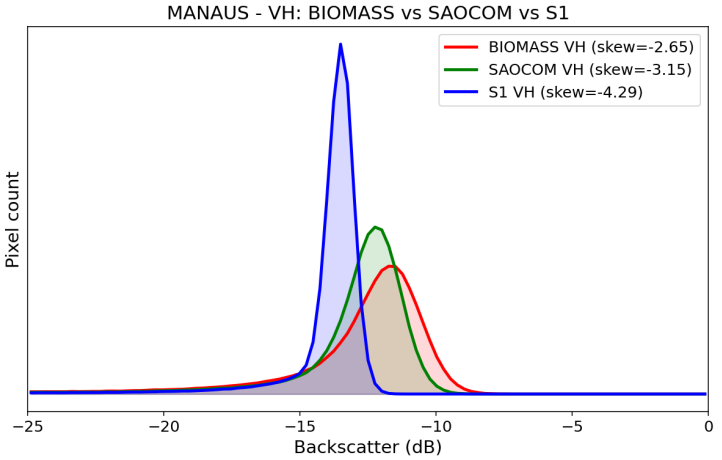
Dataset – Preliminary Analysis

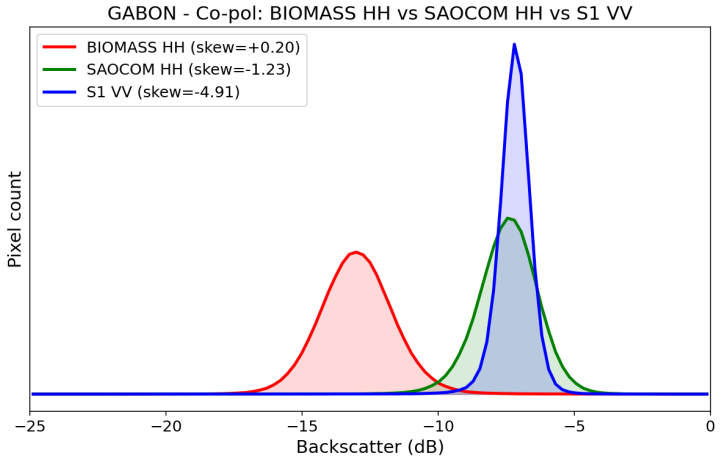
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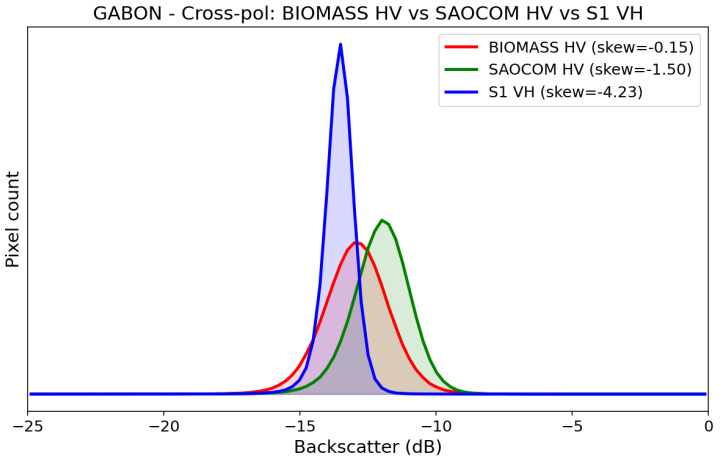
Reference data: ESA BIOMASS CCI AGB 2024





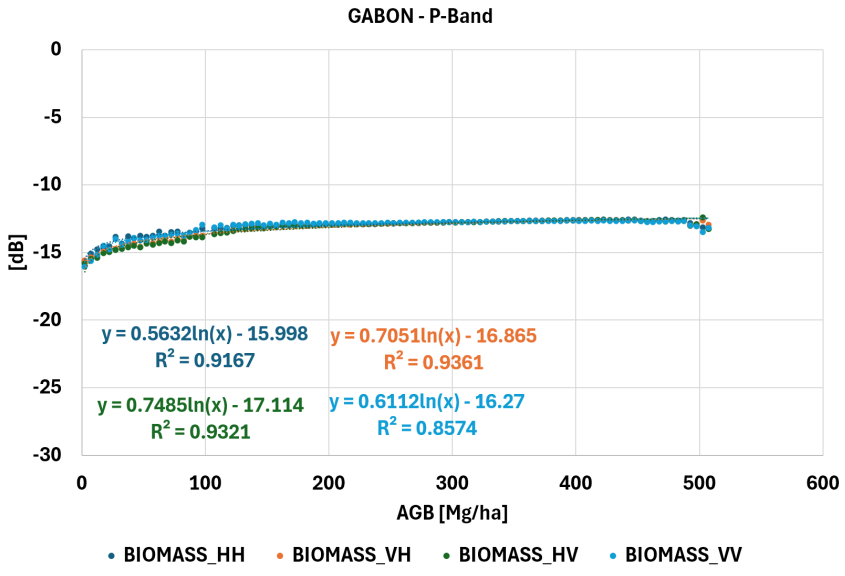


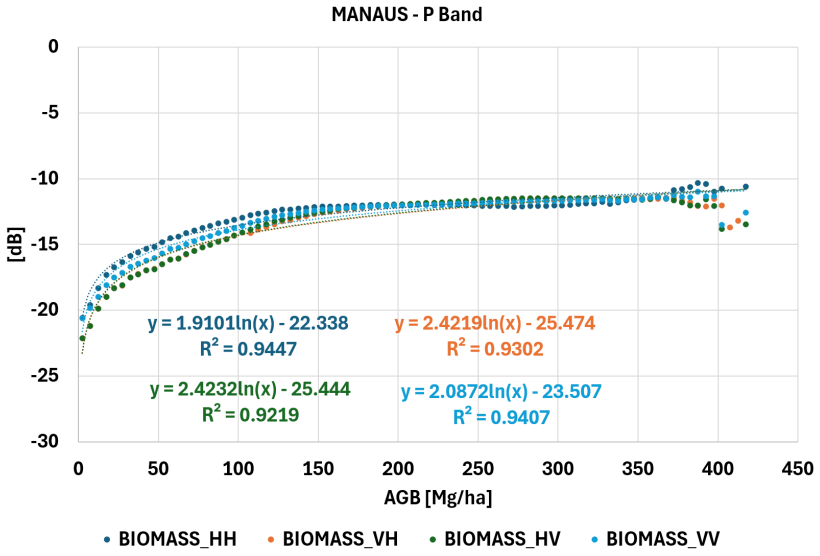


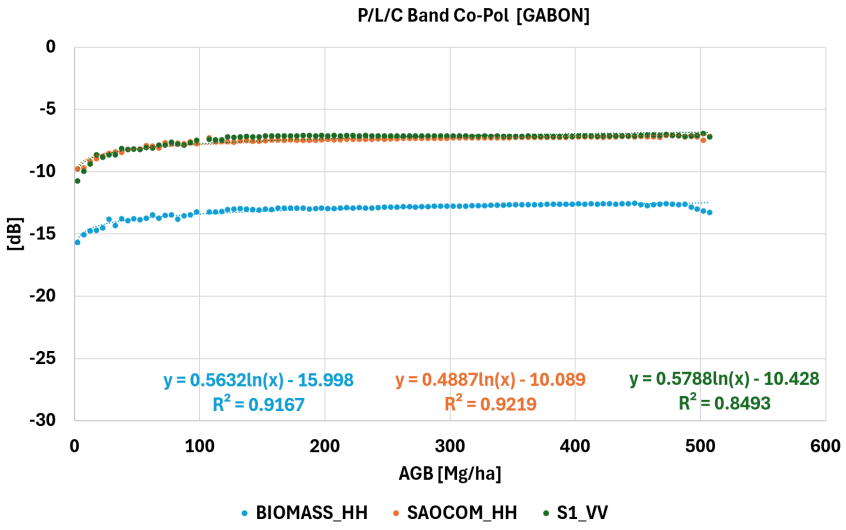


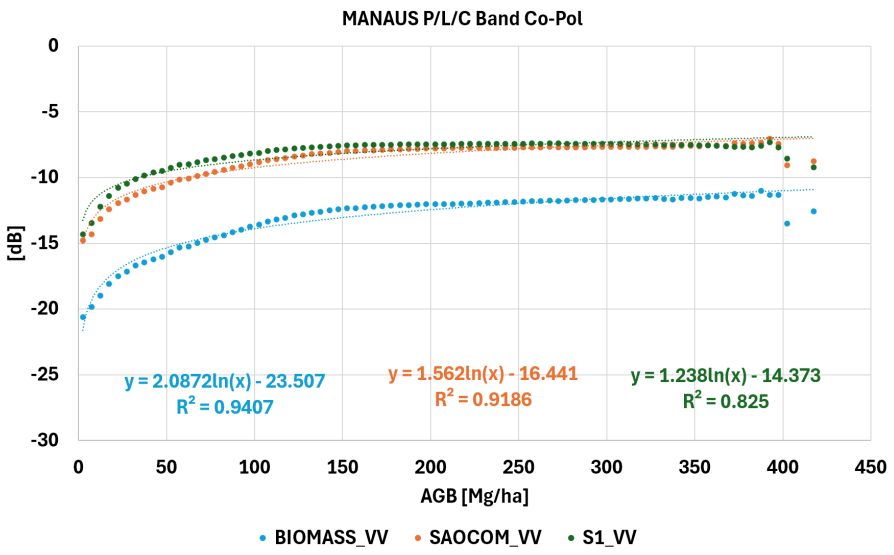
P/L/C-band backscatter vs AGB CCI 2024

Gabon vs Manaus - comparison across BIOMASS polarizations and AGB CCI 2024









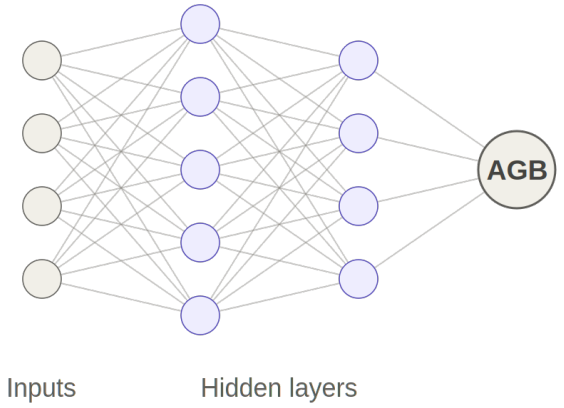
Sensitivity analysis is performed grouping the backscatter into 5 Mg/ha bins

AGB CCI is generated also by C-band Sentinel-1 monthly averages, BIOMASAR-C, ALOS-1/2, BIOMASAR-L then a weighted merge of the C- and L-band estimates

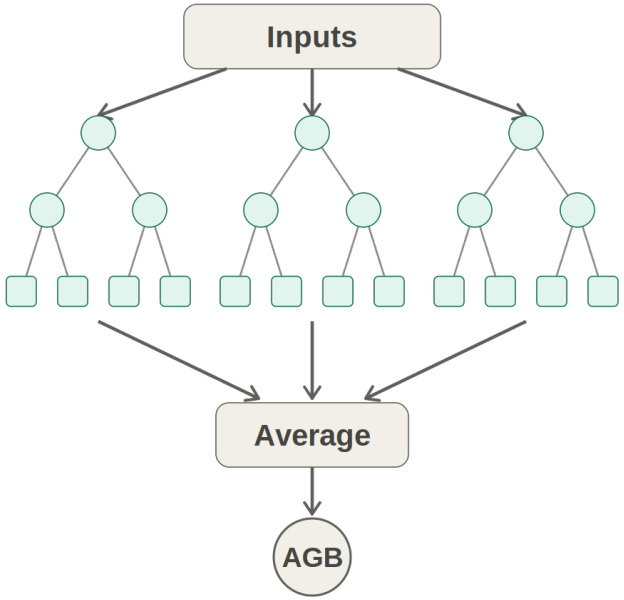
P-band responds to forest structure that the CCI map does not represent beyond saturation.

AI Algorithm: two approaches

MLP (neural network)



Random forest



Inputs:
- **SAR data (Sigma0)**
- **Local Incidence Angle**
- **DEM**
- **Slope**

Procedure for AI Algorithm implementation
Independent dataset for:
- **Training**
- **Test**
- **Validation**

AI training proxy and comparison maps
Biomass CCI, 2024
HydroGNSS Level-3 Forest Biomass (from 2026)

Work Plan

1. Data collection & sensitivity analysis

- Acquire and process BIOMASS SAR data
- Coregister auxiliary data
- Build training/validation dataset

2. AI algorithm definition

- Develop, train, test and validate inversion algorithms
- Integrate L- and C-band where available

3. AGB map generation & accuracy evaluation

- Produce AGB maps
- Compare to Level-2B AGB products
- Compare accuracy vs. HydroGNSS L3 FB